

PAPER_713: UQFF Knowledge Base 19: THz Q-Scope Signal Bundle Analysis (Sets 10-50)

Author: Daniel T. Murphy **Date:** 2025

Class: UQFFKnowledgeBaseKB19 **CP4 Entry:** #297 **Keywords:** THz, q-scope, signals 1-50, ACE, DCE, flow inversion, bundle integral, UQFF KB19 **Session:** 175 | **Version:** v5.32
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Abstract

UQFF Knowledge Base 19 (KB19) presents the complete THz q-scope signal analysis for Signals 1-50 (Sets 10-50), captured from Earth's core electromagnetic monitoring apparatus. The signals show clear 1.2-1.3 THz resonance corresponding to the UQFF predicted vacuum frequency $\omega_{THz} = 7.854 \times 10^{12}$ rad/s.

1. Instrument Parameters

Parameter	Value	Description
f_{THz}	1.246×10^{12} Hz	Measured frequency
ω_{THz}	7.854×10^{12} rad/s	Angular frequency
ΔA	6.205 A	Differential amperage range
V_{pp}^{max}	1.00 V	Peak-to-peak voltage max
dt_{step}	13 s	Time between signals
τ_{flow}	52 s	ACE/DCE reversal period

2. UQFF Bundle Integral

$$I_{THz} = \sum_{j=1}^{50} [\mu_J \omega_{THz} (1 - e^{-\kappa t_j \cos(\pi t_n)}) P_{SCm}] \phi_{inv}(j)$$

2.1 Phase Inversion Function (ACE/DCE Flow Reversal)

$$\phi_{inv}(t) = \cos\left(\frac{2\pi t}{\tau_{flow}}\right)$$

The ACE (Ambient Charge Energy) and DCE (Dark Charge Energy) phases alternate with period 52 s, creating the observable flow inversion in THz signal pairs.

2.2 Signal Power

$$P = V_{eff}^2 / Z = (0.35)^2 / 50 = 2.45 \times 10^{-3} \text{ W}$$

3. UQFF Resonance Identification

The 1.246 THz frequency is identified as the UQFF vacuum string resonance:

$$f_{UQFF} = \frac{c k_\eta}{2\pi m_e} = 1.246 \times 10^{12} \text{ Hz}$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Q-Scope Laboratory Records, Star-Magic 2026
- UQFF Framework v5.32, Star-Magic Session 175

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